Prediction of Athlete Fatigue Risk Using Machine Learning Methods

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***Abstract*—This project studies the prediction of athlete fatigue risk using supervised machine learning methods. The AFR-1000 Athlete Fatigue Risk Dataset was used as a synthetic dataset containing physiological, training-related and recovery-related variables. The target variable was `Fatigue_Risk`, which defines a binary classification task with two classes: normal and fatigued. Logistic Regression, k-Nearest Neighbors, Support Vector Machine, Random Forest and Multi-Layer Perceptron were compared against a dummy baseline. The final model was selected strictly according to the mean outer cross-validation F1-score in nested cross-validation. Test-set metrics were used only for the final independent evaluation after model selection. Logistic Regression achieved the highest mean outer cross-validation F1-score of 0.983 and was therefore selected as the final model. On the held-out test set, it reached an F1-score of 0.985. The results suggest that the synthetic fatigue-risk labels are largely linearly separable. However, because the dataset is synthetic, the results should be interpreted mainly as a comparison of machine learning methods rather than as direct clinical evidence about real athletes.**

***Index Terms*—machine learning, binary classification, athlete fatigue, fatigue risk, logistic regression, support vector machine, random forest, neural network**

I. Introduction

Athlete fatigue is an important topic in sports science because it can influence performance, recovery, training adaptation and injury risk. Fatigue is usually not caused by a single factor, but by the interaction of training load, physiological stress, recovery quality and subjective perception of effort. Monitoring training load can help coaches and athletes understand whether an athlete is adapting to a training program and may reduce the risk of non-functional overreaching, illness or injury [1]. In more severe cases, insufficient recovery after excessive training may contribute to overtraining syndrome, which is described as a maladaptive response involving several body systems [2].

Machine learning methods are suitable for this type of problem because they can model relationships between multiple input variables and a target outcome. In this project, the problem is formulated as a binary classification task. The goal is to predict whether an athlete belongs to the normal class or to the fatigued class based on demographic, physiological, training-related and recovery-related features.

The project uses the AFR-1000 Athlete Fatigue Risk Dataset [3]. The dataset is synthetic and was designed for machine learning research in sports science and athlete health monitoring. Therefore, the results should not be interpreted as direct evidence about real athletes. Instead, the dataset is used as a controlled benchmark for comparing supervised classification methods.

II. Dataset and Preprocessing Description

The AFR-1000 dataset contains 1000 samples and 15 input features. The target variable is `Fatigue_Risk`, where value 0 represents a normal state and value 1 represents a fatigued state. The input variables include demographic information, training load, physiological monitoring variables, subjective assessments and recovery indicators.

The dataset contains the following features: `Age`, `Gender`, `Training_Hours`, `Training_Intensity`, `Resting_HeartRate`, `HeartRate_Recovery`, `Lactate_Level`, `RPE`, `Sleep_Hours`, `Sleep_Quality`, `Muscle_Soreness`, `Hydration_Level`, `Cortisol_Level`, `HRV` and `Protein_Intake`.

The target variable was almost balanced: 511 samples belonged to the normal class and 489 samples belonged to the fatigued class. Therefore, no resampling method was necessary. The dataset was checked for missing values and no missing values were found.

The dataset was split into training and testing parts using stratified sampling. Stratification was used to preserve the class distribution of the target variable in both parts. The test set contained 20% of the data. For models sensitive to feature scaling, namely Logistic Regression, k-Nearest Neighbors, Support Vector Machine and Multi-Layer Perceptron, standardization was applied using `StandardScaler`. The scaler was included inside the machine learning pipeline to avoid data leakage from the test set into the training process.

III. Used Methods

Five supervised classification methods were compared in this project: Logistic Regression, k-Nearest Neighbors, Support Vector Machine, Random Forest and Multi-Layer Perceptron. A `DummyClassifier` with the most-frequent-class strategy was also included as a simple baseline.

A. Logistic Regression

Logistic Regression was used as a simple linear model for binary classification. Although the method contains the word regression in its name, it was used here as a classification algorithm. Its purpose was to test whether the two classes can be separated using a linear decision boundary. The regularization parameter C was selected using cross-validation.

B. k-Nearest Neighbors

The k-Nearest Neighbors classifier is a distance-based method. It classifies a new sample according to the labels of the closest training samples. Since KNN is sensitive to the scale of input variables, standardization was applied before training. The number of neighbors, weighting strategy and distance metric were selected using grid search.

C. Support Vector Machine

Support Vector Machine was used as a margin-based classification method. Both linear and radial basis function kernels were tested. The parameters C and γ were selected using cross-validation.

D. Random Forest

Random Forest was used as an ensemble tree-based model. It combines multiple decision trees and reduces the risk of overfitting compared with a single decision tree. The number of trees, maximum depth and splitting parameters were selected using grid search.

E. Multi-Layer Perceptron

A Multi-Layer Perceptron was used as a neural network classifier. Several network architectures and activation functions were tested. Standardization was applied before training. Early stopping was used to reduce the risk of overfitting.

F. Validation and Evaluation

Model comparison was performed using nested stratified five-fold cross-validation on the training set. In each outer fold, hyperparameters were tuned by an inner five-fold cross-validation, while the outer fold was used to estimate model performance. After the best algorithm had been selected according to the mean outer F1-score, its hyperparameters were tuned once more on the full training set and the final model was evaluated once on the independent held-out test set. The test set was used only for final evaluation and post-hoc interpretation, not for model selection, hyperparameter tuning or feature selection. Accuracy, balanced accuracy, precision, recall and F1-score were reported. F1-score was used as the main metric because it balances precision and recall in fatigue-risk detection. All random operations were controlled using a fixed random seed of 1 to improve reproducibility. All methods were implemented using scikit-learn [4].

IV. Results and Discussion

The model-selection results are shown in Table I. The models are ranked by mean outer cross-validation F1-score, because this was the criterion used for selecting the final model. Logistic Regression achieved the highest mean outer cross-validation F1-score of 0.983 with a standard deviation of 0.013 and was therefore selected as the final model. Support Vector Machine also achieved a similar outer cross-validation F1-score of 0.978, but with a higher standard deviation across folds. Multi-Layer Perceptron also achieved strong cross-validation performance, while Random Forest and k-Nearest Neighbors performed worse. The dummy baseline achieved a cross-validation F1-score of 0 because it always predicted the most frequent class and therefore did not identify the fatigued class.

TABLE I
Model comparison using nested cross-validation.

Model	Outer CV F1-score	Outer CV accuracy
Logistic Regression	0.983 ± 0.013	0.984 ± 0.012
SVM	0.978 ± 0.017	0.979 ± 0.017
MLP	0.966 ± 0.008	0.968 ± 0.007
KNN	0.904 ± 0.027	0.909 ± 0.025
Random Forest	0.884 ± 0.012	0.889 ± 0.011
Dummy Classifier	0.000 ± 0.000	0.511 ± 0.002

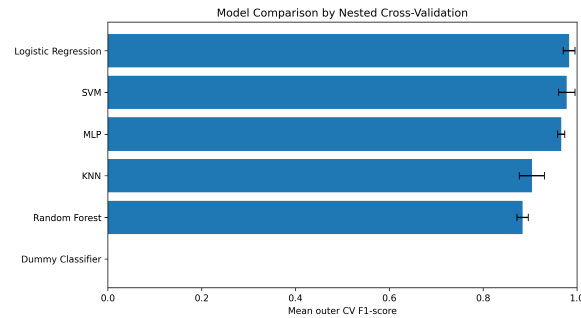

Fig. 1. Model selection according to mean outer cross-validation F1-score. Error bars show the standard deviation across folds.

Figure 1 shows the same model comparison based on outer cross-validation F1-score, including the standard deviation across folds.

The high performance of Logistic Regression and Support Vector Machine suggests that the labels are largely linearly separable. This is consistent with the description of the dataset, where labels were generated according to physiological logic linking training load, recovery and stress-related indicators to fatigue risk. Logistic Regression was selected because it achieved the highest mean outer cross-validation F1-score with low variability.

After the final model had been selected by cross-validation, it was evaluated on the independent test set. The confusion matrix of the selected model is shown in Fig. 2. Logistic Regression made only three incorrect predictions on the test set: two normal samples were classified as fatigued and one fatigued sample was classified as normal. The final test metrics are summarized in Table II. They confirm the strong performance of the selected model on the held-out test set.

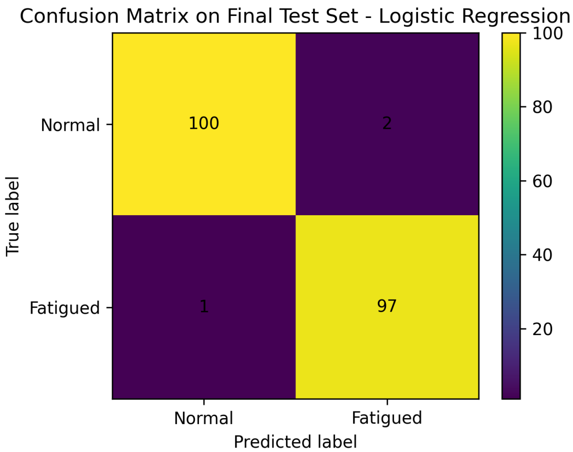

Fig. 2. Confusion matrix of the selected Logistic Regression model on the held-out test set.

TABLE II
FINAL HELD-OUT TEST EVALUATION OF THE SELECTED LOGISTIC REGRESSION MODEL.

Metric	Value
Accuracy	0.985
Balanced accuracy	0.985
Precision	0.980
Recall	0.990
F1-score	0.985

A. Permutation Importance

Permutation importance was used to interpret the selected Logistic Regression model more explicitly. It was computed on the held-out test set after the final model had already been selected. The analysis was used only for post-hoc interpretation and was not used for feature selection, model tuning or additional model comparison. The method measures how much the F1-score decreases when one feature is randomly shuffled while all other features remain unchanged. A larger decrease means that the selected model depends more strongly on that feature for correct predictions on unseen data.

The most important features are listed in Table III and visualized in Fig. 3. The highest permutation importance values were obtained for `HRV`, `Cortisol_Level`, `Sleep_Quality`, `HeartRate_Recovery`, `RPE` and `Muscle_Soreness`. These results suggest that fatigue-risk predictions in the dataset were mainly associated with recovery-related and stress-related indicators.

TABLE III
TOP FEATURES ACCORDING TO PERMUTATION IMPORTANCE OF THE SELECTED LOGISTIC REGRESSION MODEL.

Feature	Mean decrease in F1-score
HRV	0.259
Cortisol_Level	0.157
Sleep_Quality	0.151
HeartRate_Recovery	0.148
RPE	0.114
Muscle_Soreness	0.107
Training_Intensity	0.087
Sleep_Hours	0.085
Lactate_Level	0.068
Training_Hours	0.051

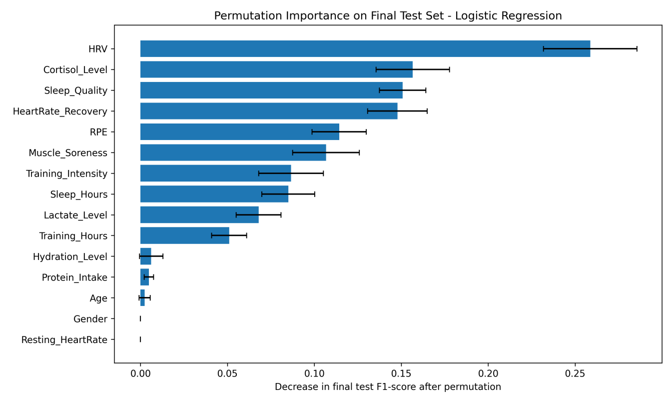

Fig. 3. Permutation importance of the selected Logistic Regression model on the held-out test set. Higher values mean a larger decrease in F1-score after feature permutation.

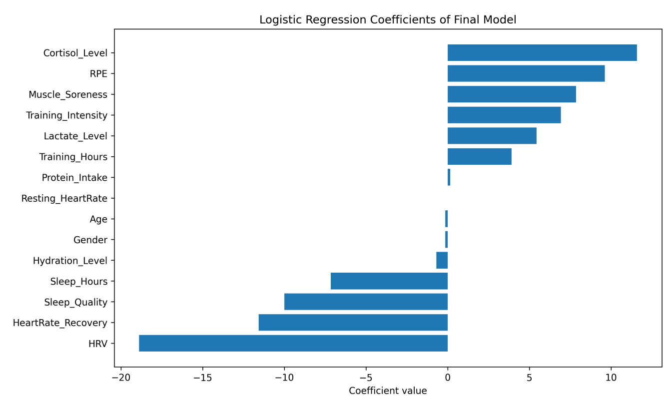

Fig. 4. Coefficients of the selected Logistic Regression model. Positive coefficients increase the predicted probability of fatigue, while negative coefficients decrease it.

The Logistic Regression coefficients in Fig. 4 show a similar pattern. Higher HRV, better heart rate recovery, better sleep quality and longer sleep duration were associated with a lower predicted probability of fatigue. In contrast, higher cortisol level, higher perceived exertion, higher muscle soreness, higher training intensity and higher lactate level were associated with a higher predicted probability of fatigue. Because the model was trained in a pipeline with feature standardization, the coefficient magnitudes are more comparable across input variables than they would be on the original feature scales.

Heart rate variability, abbreviated as HRV, describes variation in time intervals between consecutive heartbeats. HRV is commonly used as a non-invasive indicator of autonomic regulation and has been applied in sport and strength-and-conditioning contexts to assess training status, adaptability and recovery. In general, higher HRV is associated with better recovery and lower physiological stress, while reduced HRV may indicate accumulated fatigue or insufficient recovery [5]. Therefore, the negative coefficient of HRV in the Logistic Regression model is consistent with the expected interpretation: higher HRV decreases the predicted probability of fatigue risk.

However, the results must be interpreted carefully. The dataset is synthetic, and therefore the high performance does not prove that fatigue risk in real athletes can be predicted with the same accuracy. The models most likely recovered the decision logic used during synthetic data generation. Real athlete data may contain measurement noise, individual variability, temporal dependencies and other factors that are not fully captured in this dataset.

V. Conclusion

This project compared several supervised machine learning methods for predicting athlete fatigue risk using the AFR-1000 synthetic dataset. The problem was formulated as a binary classification task. Model selection and hyperparameter tuning were performed only on the training data using nested stratified cross-validation. The inner loop was used for hyperparameter tuning, while the outer loop was used to estimate model performance. The best model according to mean outer cross-validation F1-score was Logistic Regression, which achieved a mean outer cross-validation F1-score of 0.983. This outer cross-validation score, not the held-out test score, was the basis for selecting the final model.

After model selection, the selected Logistic Regression model was evaluated on the held-out test set and achieved an F1-score of 0.985. The confusion matrix showed only three misclassified test samples. Permutation importance was then computed on the test set only as a post-hoc interpretation method. It was not used for feature selection or further model tuning.

The results suggest that a simple linear model is sufficient for this dataset. More flexible models such as SVM, Random Forest and Multi-Layer Perceptron did not clearly outperform Logistic Regression. This is probably caused by the synthetic nature of the dataset and the fact that the target variable was generated according to predefined physiological rules. The obtained scores are therefore unusually high compared with what would typically be expected in real-world fatigue prediction. Since such high performance can easily be overestimated if information from validation or test data leaks into training, the evaluation procedure was designed to reduce this risk. All preprocessing steps were performed inside scikit-learn pipelines, hyperparameter tuning was restricted to the training folds within nested cross-validation, and the held-out test set was used only once for the final evaluation.

Future work should focus on validation using real athlete data. It would also be useful to include temporal information, because fatigue often develops over time and may depend on previous training sessions, recovery history and long-term trends.

References

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